

Understanding of sailing rule based on COLREGs: Comparison of navigator survey and automated collision-avoidance algorithm

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ABSTRACT

The Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) comprises rules to prevent collisions at sea and is based on the qualitative rules and ordinary practice of seamen. However, because the criteria of the sailing rule interpretation are different, the problem of misunderstanding occurs owing to the difference in the interpretation between navigators and collision-avoidance algorithm designers. Therefore, this study aims to clarify the collision situations by understanding in the sailing rule interpretation of COLREGs. To identify the understanding of collision situations, we surveyed the navigators' understanding of collision situations and reviewed the collision-avoidance algorithms used in recent studies. The survey results were analyzed based on head-on and crossing (HC) and crossing and overtaking (OC) situations. Results showed that navigators were unsure about whether the HC situation sailing rule should be applied to incoming vessels from the 008° and whether the overtaking or crossing situation should be applied to coming vessels at 160°. In OC situations, as the angle increased from 130°, it became difficult for navigators to interpret the navigation rules. A notable disparity emerged between the navigators' understanding and the automated collision-avoidance algorithms. Moreover, it was confirmed that practical navigators exhibited divergent interpretations of these regulations. This study contributes to provide an understanding of COLREGs sailing rules based on the understanding of navigators and researchers.

1. Introduction

Ship collision accidents are an important factor that threaten the safety of marine navigation [54]. Moreover, they lead to serious issues such as environmental pollution, human casualties, and economic losses [55,77]. The International Maritime Organization (IMO) created the Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) in 1972 to prevent ship collisions and ensure safe passage of ships. COLREGs came into effect in 1977 [30] and comprises the qualitative rules and ordinary practice of seamen [52,72]; as such, COLREGs is rules at sea for the safe navigation of vessels by avoiding risks based on the decision-making by seamen [79].

Despite the aforementioned rules, marine accidents of vessels have occurred steadily every year [28,26]. Human factors have highly contributed to collision accidents [74,81], and various efforts have been made to prevent accidents caused by seamen [6,56,69,83]. In case of merchant ships, human error is the most important factor that directly affects the safety and security at sea [60] and is caused by the lack of

training and heavy workload [80,53]. To address this problem, a study was conducted on the reduction of the causes that generate navigator errors using sensors [84]. Furthermore, ship automation has emerged as a solution to fundamentally reduce the causes of marine accidents [19,35,37].

Owing to the aforementioned developments, existing ship operations based on navigators have been gradually changed to autonomous ships in recent years [1,19,48,58]. In the past, navigators have operated a ship through their decision-making, which has changed to ship operations through systems and sensors with the emergence of maritime autonomous surface ships (MASSs) that can operate autonomously [4,75]. MASSs have been developed using advanced technologies during the Fourth Industrial Revolution by many nations and institutions [32], and studies at a working-level stage have been reported in recent years [68]. Autonomous navigation is difficult to be applied to all ships at the current technical stage; nevertheless, the construction of the first MASS began in Norway in 2020 as an unmanned electric cargo ship [94]. In Japan, a report announced that autonomous navigation tests were

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conducted along a 300-km-long sea route using a 194-twenty-foot-equivalent-unit container ship in 2022; this ship achieved autonomous navigation without boarding crew. South Korea also reported a successful demonstration of fully autonomous navigation of a 12-passenger cruise ship in 2021. These achievements indicate the emergence of real MASSs at sea.

However, the emergence of autonomous vessels has created a new problem in that navigators should now consider differences in navigations between manned ships and machine-operated unmanned ships in addition to existing ship navigations operated only by navigators [47, 64,71]. The navigations between manned ships and machine-operated unmanned ships are considered as an additional variable compared with existing marine transportation, which considers only manned ships. The IMO has categorized the development of MASSs into four degrees of autonomy [29,32]: Degree one: Ship with automated processes and decision support, Degree 2: Remotely controlled ship with seafarers on board, Degree 3: Remotely controlled ship without seafarers on board, and Degree 4: Fully autonomous ship ([13,33,16]). However, Degrees 2–4 have presented problems. If a same collision-avoidance model can be applied to unmanned and manned ships, safe passage can be ensured; otherwise, there is always a risk of collision at sea. Therefore, several studies have developed a collision-avoidance model that can be applied to both manned and unmanned ships based on COLREGs.

COLREGs categorize the risks of colliding ships into three models, as shown in Fig. 1 [30]. The three models include COLREG Rule 13. Overtaking, Rule 14. Head-on, and Rule 15. Crossing Situation: Give-way vessel and stand-on vessel. Each of these models performs a collision avoidance action based on the relative orientation of the other ship; moreover, these models are expected to perform a collaborative action. Machines will perform an algorithm according to COLREGs, and navigators will also perform the avoidance action according to the three models in COLREGs [2]. The collision-avoidance model can be clearly applied to places that are not in the vicinity of the boundaries in the three models, but “misunderstanding” occurs because the sailing rules and interpretation of the situation are needed in the vicinity. One of the representative ambiguities of COLREGs may occur when the relative bearing of the target ship is 112.5° in the overtaking or crossing courses [51]. Based on the interpretation of these circumstances, a give-way or stand-on vessel can be determined. Thus, the collision avoidance method is determined on the basis of the interpretation.

In summary, navigators make a decision according to the collision-avoidance model of COLREGs and the basis of the decision should follow the situations shown in Fig. 1. Based on this, automatic avoidance algorithms of MASSs in recent years are based on the collision-avoidance

model of COLREGs, and these algorithms or systems must strictly follow these rules. Furthermore, a question arises about how accurately the navigators understand COLREGs and interpret rule requirements. This question arises because COLREGs is based on qualitative rule and ordinary practice seamanship and not because navigators do not understand these rules [22]. The current COLREGs clearly defines the overtaking situation. This is not the case for but crossing to head-on or overtaking situations. Therefore, this study aims to clarify the collision situations by understanding the sailing rule interpretation of COLREGs.

In this study, we identified the recognition of crossing situations to verify the understanding of collision avoidance situations in COLREGs. We performed a comparative analysis between the autonomous collision-avoidance algorithm and the navigator's perception of the crossing situation bearing from own ship; subsequently, we analyzed the collision situations between manned and unmanned ships. In Section 2, we present the study methodology, in which the provisions and survey method are described in relation to collision defined in COLREGs. In addition, the collision-avoidance models used in current studies are reviewed and accident cases are discussed. In Section 3, collision avoidance actions performed by navigators are discussed. In Section 4, the results of the investigation are analyzed. In Section 5, the collision situations based on the results and the boundary of bearing from own ship are discussed. Finally, conclusions are described in Section 6.

2. Methodology

This study aimed to evaluate the navigators' understanding of COLREGs during crossing situations; therefore, we first performed a review on the COLREGs rules [30]. In addition, we reviewed the rules applied to the collision avoidance algorithm, which has been used in unmanned MASSs in recent studies. We recognized accidents caused by misunderstanding in sailing situations of collision accidents that occurred between manned ships [36] and defined the collision avoidance situation bearing from own ship. Subsequently, we performed a survey with navigators to collect the sailing rule interpretation data about crossing situations based on COLREGs. The survey questionnaires were based on COLREGs and radar screens that can generally recognize the collision avoidance situations of other ships identified by the navigators. Furthermore, the results are presented with quantitative data and the recognition of situations by the navigators is interpreted with respect to social science and engineering.

2.1. COLREGs

Collision rules have gained importance as ships have employed engines since the Industrial Revolution [11]. This has led the IMO to revise COLREGs to the latest revision at an international conference held in London in 1972 to prevent collisions at sea [30]. Navigators who perform navigation commonly agree that a clear understanding of the provisions of COLREGs is crucial for preventing collision accidents to realize MASSs [52].

COLREGs have been written with general descriptions to be applied to as many situations as possible [12,87] that require decision-making based on the experience of navigators and maritime culture [6]. Thus, we need to check the elements where these vague situations occur by using case studies of real accidents and literature reviews on these three collision-avoidance model situations [73]. The three collision-avoidance models of COLREGs follow Rule 13, overtaking Rule 14. Head-on situation, Rule 15. Crossing situation as shown in Fig. 2. Based on these models, the role of Rule 17. Give-way vessel and Rule 18. Stand-on vessel is divided into sectors.

2.1.1. Rule 13, overtaking

(a) Notwithstanding anything contained in the rules of part B, Sections I and II any vessel overtaking any other shall keep out of the way of the vessel being overtaken.

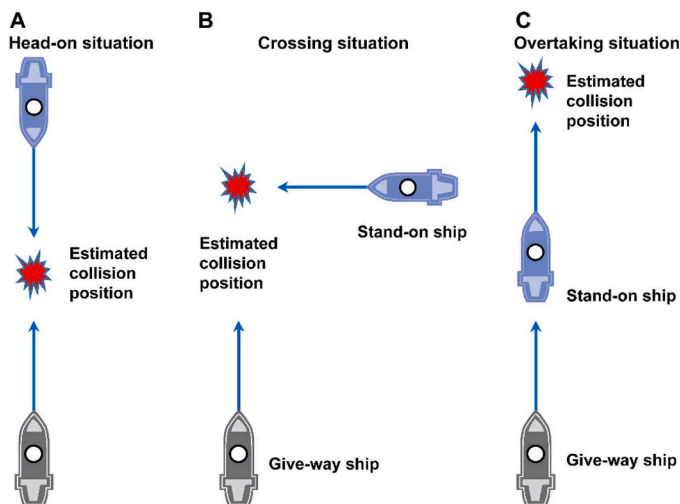


Fig. 1. Navigation models according to COLREGs.

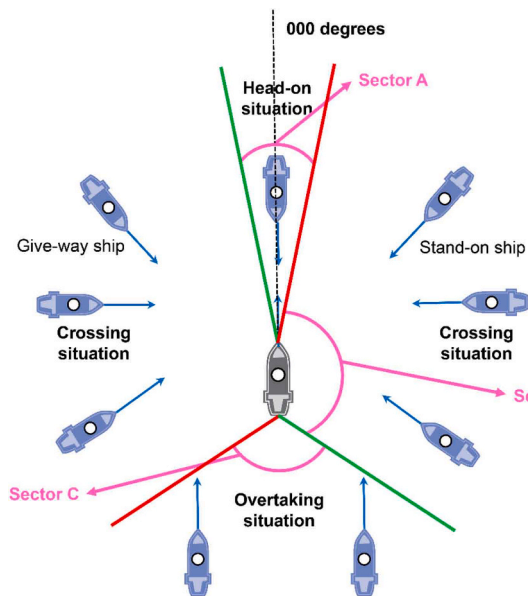


Fig. 2. Navigation models according to COLREGS (Sector A refers to the angle of head-on situation, Sector B refers to the angle of crossing situation, and Sector C refers to the angle of Overtaking situation.).

(b) A vessel shall be deemed to be overtaking when coming up with another vessel from a direction more than 22.5° abaft its beam, i.e., in such a position with reference to the vessel but neither of its sidelights.

(c) When a vessel is in any doubt as to whether it is overtaking another, it shall assume that this is the case and act accordingly.

(d) Any subsequent alteration of the bearing between the two vessels shall not make the overtaking vessel a crossing vessel within the meaning of these rules or relieve it of the duty of keeping clear of the overtaken vessel until it is finally past and clear.

2.1.2. Rule 14, head-on situation

(a) When two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter its course to starboard so that each shall pass on the port side of the other.

(b) Such a situation shall be deemed to exist when a vessel sees the other ahead or nearly ahead and by night it could see the masthead lights of the other in a line or nearly in a line and/or both sidelights and by day it observes the corresponding aspect of the other vessel.

(c) When a vessel is in any doubt as to whether such a situation exists, it shall assume that it does exist and act accordingly.

2.1.3. Rule 15, crossing situation

When two power-driven vessels are crossing so as to involve risk of collision, the vessel, which has the other on its own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.

2.1.4. Rule 16, action by give-way vessel

Every vessel, which is directed by these rules to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear.

2.1.5. Rule 17, action by stand-on vessel

(a) (i) Where one of two vessels is to keep out of the way the other shall keep her course and speed.

(ii) The latter vessel may however take action to avoid collision by its maneuver alone, as soon as it becomes apparent to it that the vessel required to keep out of the way is not taking appropriate action in compliance with these rules.

(b) When, from any cause, the vessel required to keep her course and speed finds itself so close that collision cannot be avoided by the action of the give-way vessel alone, it shall take such action as will best aid to avoid collision.

2.1.6. Misunderstanding of sailing rule

Misunderstanding occurs owing to the expression “in any doubt as to whether such a situation she shall act accordingly” during the head-on and crossing situation and overtaking and crossing situation in Rules 13 (c) and 14 (c) of COLREGs. The current COLREGs clearly defines the overtaking situation. However, the relationship between head-on and crossing as well as crossing and overtaking requires requirement rather than clear definition. Even the stand-on vessel in Rules 16 and 17 is not completely exonerated from collision avoidance, and misunderstanding of collision avoidance time also occurs. Thus, how researchers apply navigation in collision situations in compliance with COLREGs should be reviewed and based on this review results, survey questionnaires should be configured.

2.2. Collision-avoidance algorithm for MASSs

The application of the collision avoidance algorithm for MASSs is mainly divided based on whether COLREGs are considered or not [3,8,24]. In addition, collision avoidance is divided into safe distance and trajectory prediction methods, and these methods are also based on the compliance of the sailing rules described in COLREGs as the basic principle [39]. When COLREGs is considered, situations are divided into three collision avoidance algorithms [70]. Even if COLREGs is not considered, the distance to a ship and evaluation of collision risk are used as collision avoidance methods, indicating that COLREGs is indeed considered ([40,78,89]). Appendix 1 presents the related studies on collision avoidance of a ship by applying COLREGs. When COLREGs was considered, the angles in the three collision avoidance models were divided. In the head-on situation, the angles from own ship considered were distributed from 0° to 50° and 10° was the most considered angle. In the crossing situation, a range of angles from own ship was 90° – 170° , in which 112.5° was mostly used in recently reviewed studies. Finally, in the overtaking situation, a range of angles from own ship was 20° – 180° from the stern and 135° was the most used angle in recent studies. In other studies, COLREGs was applied, but they did not describe accurate angles. Thus, these studies were not included in this literature review.

2.3. Marine accident cases due to misunderstanding of collision situations

There are various cases of vagueness situations upon collision situations of COLREGs out of marine accidents at sea in Korea. We investigated many marine accidents for the last 10 years in the Korea Maritime Safety Tribunal, an agency that investigates maritime accidents [36], and extracted case that best represented the misunderstanding of collision situations and performed a case study: **Collision accident between product tanker DL Sunflower and bulk carrier Ligari**.

There was a collision between the bulk carrier and oil product tanker in the South Sea of South Korea at approximately 02:30 am on January 11, 2014. The gross tonnage of the tanker was 28,519 tons and the port of registry was Jeju-si. It was built and launched for the first time in Japan in 1998. The bulk carrier was the Maltese registered ship whose gross tonnage was 38,851 tons. It was built and launched for the first time in Japan. The tanker departed from Ulsan Port and sailed a true course of 206° for the Philippines at a speed of ~ 12.5 knots at approximately 02:00 am, which was 30 min before the collision. At the same time, the bulk carrier departed from the Port of Vancouver USA and sailed a true course of 235° for the Port of Longkou in China at a speed of ~ 11.4 knots. As shown in Fig. 3, the two ships were two miles apart and DL Sunflower found Ligari on its portside. The angle of direction is 250° from Sunflower Bow. The two ships agreed on how to

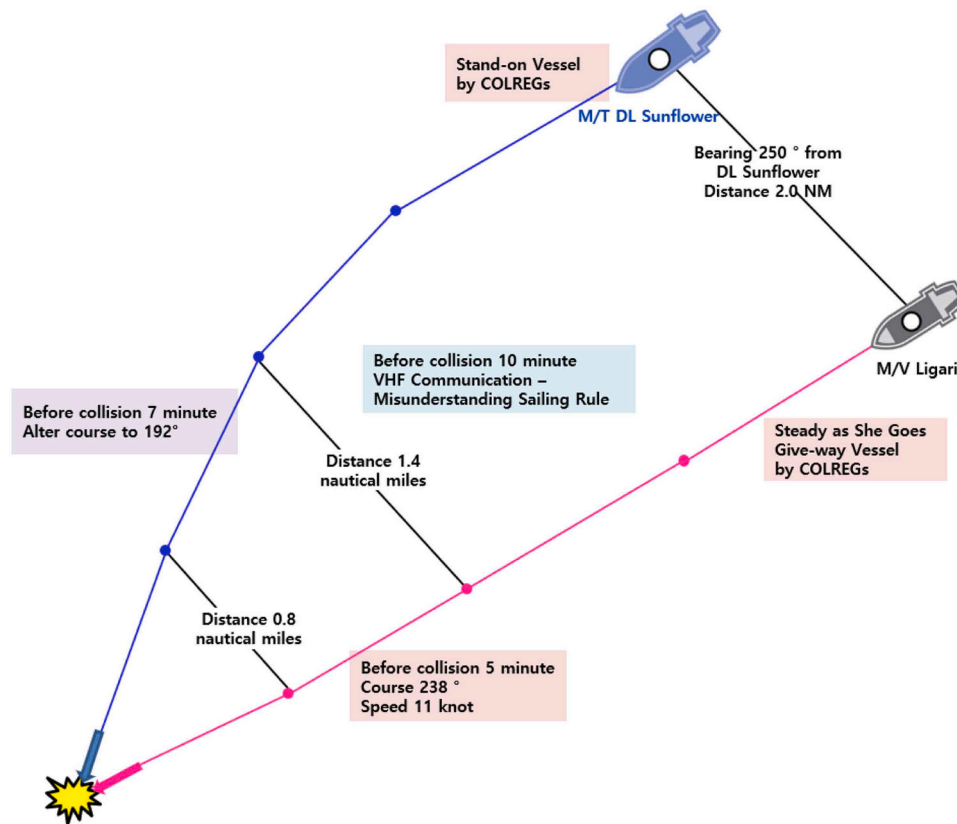


Fig. 3. Marine collision accident due to the misinterpretation of the sailing rule.

avoid collision 10 min before the collision while they met in a crossing situation. The second mate of that tanker thought its ship would pass around the stern of the bulk carrier under the condition that the bulk carrier would maintain its course as a give-way ship and alter the course to starboard. Furthermore, the second mate of the bulk carrier thought the agreement was made because tanker preferred to sail south. Both ships did not correctly identify the risk. In particular, when a call was made from the tanker via VHF 10 min before the collision, the bulk carrier reached an agreement without correctly knowing the collision avoidance navigation with the tanker. The misunderstanding of the collision situation based on COLREGs was not identified. Consequently, the collision was not avoided without taking collision avoidance action according to the proactive and proper seamanship with sufficient time to take action as a result of the incorrect interpretation of navigation.

2.4. Clear understanding for collision avoidance situations

COLREGs is implicitly assumed to completely cover all collision prevention requirements. Currently, COLREG is deliberately compiled [35,37]. In Rules 13–15, situations are clearly defined and practicable on board a ship, particularly via compass investigation. For the purpose of safety, an item of the provisions will tackle problems, such as “When a vessel is in any doubt as to whether such a situation exists, she shall assume that it does exist and act accordingly.” The misunderstandings about the collision situation were confirmed to occur owing to the rule’s intention and coverage. This intentional misunderstanding allows the free interpretation of wide arrangements in complex collision prevention scenarios without excessive limitation for designers of MASS collision prevention, as presented in Appendix 1 [22,52,72]. However, the difference bearing from own ship between the researchers of collision-avoidance algorithm and the sailing rule interpretation of navigators is likely to lead to marine accidents [67].

COLREGs did not provide specific quantitative values for the head-on

situation. Thus, many values were applied recently, among which a range of 5–120° was used. Because 112.5° was presented in the overtaking rule, it was mostly used in the crossing situation; its range was 90–170°. Finally, 135° was the most used angle in overtaking because overtaking employed a remaining angle based on crossing, which distributed a range of 20–180°. As verified through the applied angles from the own ship, algorithms have problems due to misunderstanding resulting from the qualitative rule application of COLREGs. These problems can also be verified in accidents caused by errors in the navigation relations between manned ships.

Furthermore, in the left-crossing situation, also known as crossing, the stand-on ship is the give-way vessel and the target ship is designated as the give-way vessel [30]. In this scenario, the stand-on ship must maintain its course and speed according to the COLREGs. This situation is similar to the navigation of a stand-on ship in an overtaking situation. These findings show that considerable differences might not exist in mariners’ responses between these two situations. However, in the right-crossing situation, the stand-on ship becomes the give-way vessel and the interpretation of navigation differs in case of an overtaking situation. Consequently, mariners’ navigation interpretations might vary depending on the angle. Therefore, we focused solely on the right-crossing situation and did not consider the left-crossing situation.

We define the differences between the collision-avoidance algorithm employed based on MASS and navigators’ interpretations of sailing rules according to the situation that best allows us to verify navigators’ sailing rule interpretations. To achieve this, two situations must be categorized according to the navigation: head-on collision (HC) and overtaking and crossing (OC); ship operators must be able to interpret the sailing rules. In addition, the ranges must be set for HC and OC for comparison. Many recent studies have applied the range from 0° (front) to 20° in HC situations (Appendix 1). Additionally, based on the authors’ navigation experience, situations exceeding 20° cannot be considered HC situations. As shown in Appendix 1, the OC situations have been applied from

85° to 168°. Therefore, the ranges were set at 90–170° or 90–180°. Consequently, the range for HC through radar screens is from 0° (front) to 20°, whereas OC situations occur in the range from 90° to 180°. This process elucidates the difference in understanding collision avoidance situations between manned and unmanned vessels or between manned vessels themselves.

3. Surveying navigators to identify understanding of collision situations

The understanding of collision avoidance situations by navigators was identified by designating situations using radar moving screens, which are most widely used to verify collision situations at sea. We found it difficult to take many things into account in the survey. At first, the ship's navigation light and ship visual image were attached, but only radar screens were used because it was difficult to understand in the pilot test. The survey on understanding identification was conducted using questionnaires applying a Likert 5-scale. The survey was conducted for half a month on navigators who were navigators and pilots having experience on board ships. We aimed to identify the vagueness of collision-avoidance models recognized by navigators using a Google form. The questionnaire is shown in [Appendix 2 and 3](#).

Questions HC_Q1 to Q11 in [Appendix 2](#) refer to the request for identification of HC situations. Questions OC_Q1 to Q10 in [Appendix 3](#) refer to the request for identification of OC situations. For head-on situations, as verified in [Section 2.4](#). The responses to the questions were divided into five scales, which are presented in [Table 1](#).

4. Result and discussion

4.1. Results of responses in the survey questionnaire to identify navigator's understanding of collision avoidance situations

A survey on the navigator's understanding of the collision avoidance situations was conducted; the results from 101 respondents are presented in [Appendix 4](#). The survey required 20 days, and a revised questionnaire was used after the first survey version was revised with the aid of experienced captains. The scope and content required for the survey were freely determined through the face-to-face voluntary speech of participants and the final survey questionnaire was confirmed through strict confidentiality based on prior survey results.

We ensured that all 101 respondents fully understood the purpose of the survey before proceeding with the questionnaire. To enhance objectivity, we randomized the order of each situation presented in the survey. All respondents were well-versed in radar systems; to prevent invalid responses, we provided a sample example (Aspect 45°) as a means of pre-instruction.

In particular, this study conducted a survey based on Article 33 (Protection of Secrets) of the Statistics Act in South Korea to promote objectiveness and fairness of the survey and strictly protect the personal information of survey participants.

The results were displayed as the number of responses for each collision encountering angle with the average and standard deviation. In addition, [Appendix 5](#) summarizes the general statistics of the response

results. The results were categorized based on the ages of the respondents, vessel tonnage, position in the ship, and years of experience. In particular, the position in the ship was categorized into junior officer for second and third navigators and senior officer for a chief mate or higher. This categorization was performed for junior and senior officers to compare their awareness on collision avoidance situations. The distribution of junior and senior officers was nearly the same with 57% and 43%, respectively. Approximately 90% of the respondents had experience boarding large ships of over 5000 tons. Because most respondents were boarded in worldwide vessels, the results would not be substantially different from those obtained in case of worldwide navigators.

4.2. Result analysis

4.2.1. Analysis of the average and standard deviation of the respondents

To analyze the response results of identifying navigators' understanding of collision situations, [Figs. 4 and 5](#) depict two situations with the Likert scale of the responses using the average and standard deviations. As shown in [Fig. 4](#), the average and standard deviations were the lowest in Situation 1 (000°), which was a complete head-on from the front because the questions of head-on and crossing situations were configured from 000° to 020° in HC_Q1 to HC_Q11. The average tended to increase as the question continued from head-on to crossing situations. Interestingly, the standard deviation was not significantly different until HC_Q4 (006°), but from HC_Q5 (008°), it started to increase significantly. The question that had the largest standard deviation was HC_Q6 (010°), which indicated that navigators were not sure about which sailing rule should be applied between head-on and crossing situations for ships coming from 12° starboard from the head. This was analyzed as the typical result showing the misunderstanding of navigation between HC situations. At HC_Q7 (012°), the average of navigators' responses was close to three points, indicating "I am confused about the situation." Thus, most navigators recognized the misunderstanding of navigation in real situations. From HC_Q8 (014°), even if the vagueness of navigation interpretation was considered, the response average of "it is likely a crossing situation" or "it is certainly a crossing situation" increased. At HC_Q10 (018°), the average of responses indicated "it is likely a crossing situation," meaning that navigators were likely to determine that it was a crossing situation when ships came from 14° or larger from the head.

As shown in [Fig. 5](#), the average and standard deviation were the lowest in Overtaking OC_Q1 (000°), which was complete overtaking

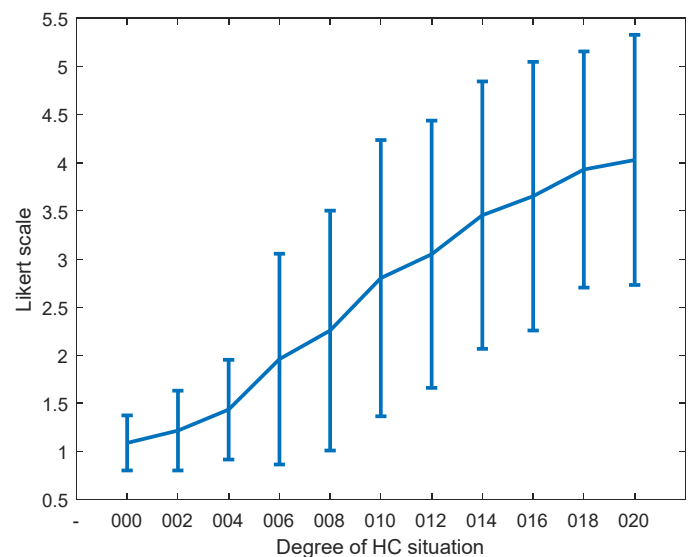


Fig. 4. Survey results of navigators' understanding of collision situations for HC_Q1 to HC_Q11.

Table 1

Survey questionnaire to identify navigator's understanding of collision avoidance situations.

Response to the questions about head-on and crossing situations (HC_Q1 to HC_Q11)	Response to the questions about overtaking and crossing situations (OC_Q1 to OC_Q10)
1. It is certainly a head-on situation.	1. It is certainly an overtaking situation.
2. It is likely a head-on situation.	2. It is likely an overtaking situation.
3. I am confused about the situation.	3. I am confused about the situation.
4. It is likely a crossing situation.	4. It is likely a crossing situation.
5. It is certainly a crossing situation.	5. It is certainly a crossing situation.

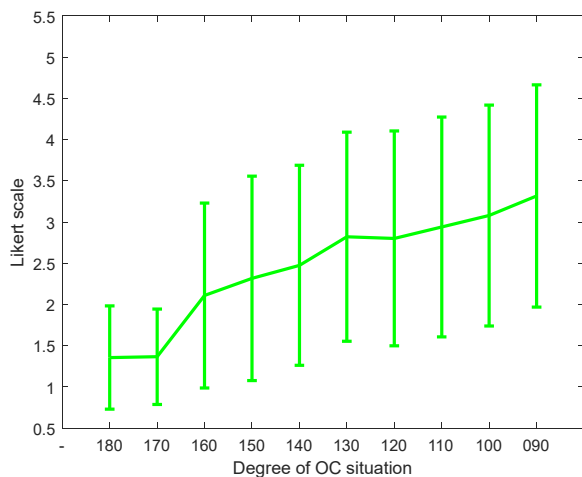


Fig. 5. Survey results of navigators' understanding of collision situations for OC_Q1 to OC_Q10.

from the stern because the questions of OC situations were configured from 180° to 090° in OC_Q1 to OC_Q10. Furthermore, the standard deviation started to increase from OC_Q3 (160°). Interestingly, as the question from OC_Q3 (160°) to OC_Q10 (090°) progressed, the standard deviation gradually increased and the question with the largest standard was OC_Q10 (090°). This indicated that navigators were uncertain about which sailing rule should be applied between OC situations for ships coming from 90° starboard from the stern. This was analyzed as a typical result that showed the misunderstanding of navigation between OC situations. In addition, most navigators recognized the vagueness of the sailing rule in real situations as the average of navigators' responses was close to three points, indicating "I am confused about the situation." from OC_Q9 (130°) and beyond. Moreover, from OC_Q10 (090°) and beyond, the average of the responses that "it is likely a crossing situation" increased even if the misunderstanding of navigation interpretation was considered, indicating that navigators determined whether the existing situation was a crossing situation when ships came from 090° or lower from the heading.

Figs. 6 and 7 show the evaluations of the questions by junior and senior officers. Generally, ship maneuvering is difficult as it requires various experiences and knowledge (Perara, 2017). Thus, senior officers

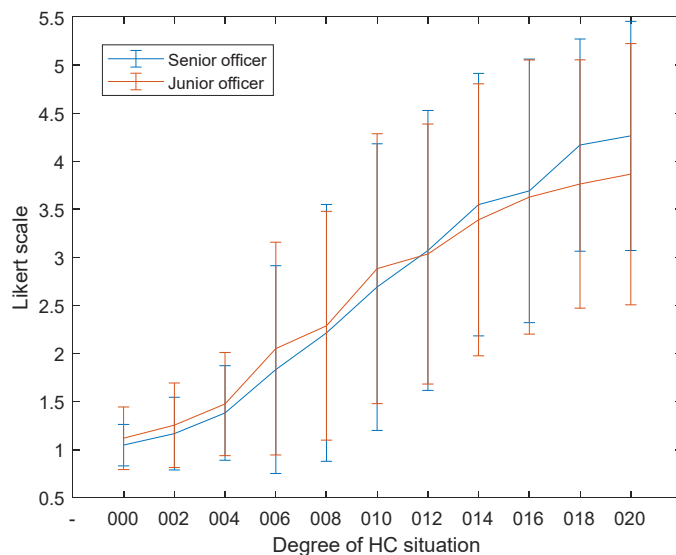


Fig. 6. Difference in the understanding of collision situation between junior and senior officers for the HC situation.

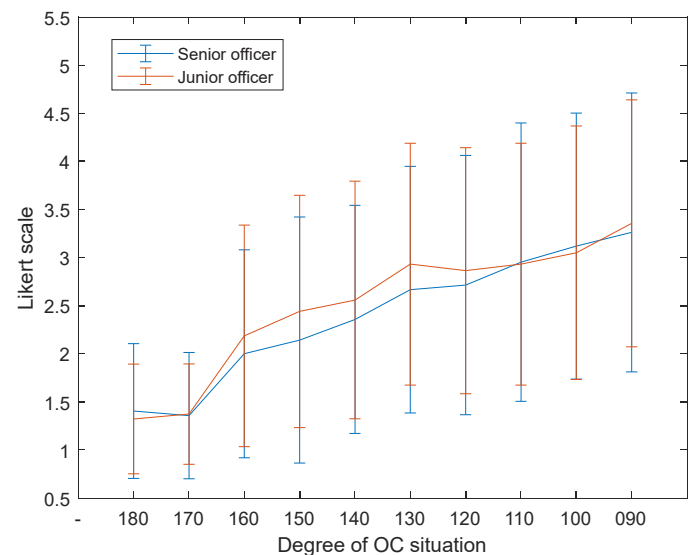


Fig. 7. Difference in the understanding of collision situations between junior and senior officers for the OC situation.

with various experiences and knowledge would have higher situation understanding. To verify this statistically, we performed comparative analysis between two groups. As shown in Fig. 6, the questions of head-on and crossing situations were configured from 000° to 020° in HC_Q1 to HC_Q11 and no significant difference in the average were observed between the two groups. At HC_Q7(012), the average was the same and junior officials had better understanding of collision situations than senior officials although the different was not significant. The standard deviation was nearly the same between the two groups, and the graph fluctuated owing to the average difference.

As shown in Fig. 7, the average and standard deviations were the lowest in case of Overtaking OC_Q1 (000°), which was complete overtaking from the ship's stern, in both groups because the questions of OC situations were configured from 180° to 090° in OC_Q1 to OC_Q10. The average scale of the junior officers was mostly lower than that of the senior officers, which is interpreted as junior officials perceiving overtaking situations more than senior officers. In addition, higher standard deviations were observed in almost all questions for junior officers than for senior officers.

4.3. Validation

To confirm the validation of the questionnaire result, factor and reliability analyses were performed. Factor analysis is a concept widely used to verify the validity of the measurement tool; the SPSS (ver. 22.0) program was used for this analysis. The results of factor analysis for each of the two situations are shown in Appendix 6 and 7.

As shown in Appendix 6, factor analysis was performed to understand how the HC factors of the encountering state are classified. As a factor extraction method, principal axis factor extraction was performed; moreover, varimax rotation was performed. Consequently, there were no factors impeding validity, and factor analysis was performed using 11 items. The KMO measure was 0.843, and the Bartlett's sphericity test result showed a significance probability of < 0.05, indicating that the factor analysis model was appropriate. Furthermore, the cumulative variance was 76.83%, indicating that the explanatory power of the four factors was high. All factor loading values were > 0.4, which satisfied the validity of the overall measurement tool; further, the analysis was conducted without additional item exclusion or adjustment.

Factor analysis was performed as shown in Appendix 7 to understand how the factors of the other participant in the overtaking state are

classified. As a factor extraction method, principal axis factor extraction was performed; moreover, varimax rotation was performed. Therefore, there were no factors impeding validity, and a factor analysis was performed using 10 items. The KMO measure was 0.859, and the Bartlett's sphericity test result showed a significance probability of < 0.05 , indicating that the factor analysis model was appropriate. Furthermore, the cumulative variance was 76.035%, indicating that the explanatory power of the three factors was high. All factor loading values were > 0.4 , which satisfied the validity of the overall measurement tool, and the analysis was conducted without additional item exclusion or adjustment.

To measure the internal consistency of the Likert-scale format survey, we used Cronbach's alpha reliability coefficient. We calculated the reliability coefficient for each question item and determined the overall reliability based on the average of the calculated reliability coefficients. A higher average correlation coefficient indicates a larger Cronbach's alpha value. Appendix 8 presents the results of the reliability analysis for each survey. Generally, a value of ≥ 0.7 indicates a strong correlation. The Cronbach's alpha coefficients calculated for the survey items for HC and OC were > 0.7 , demonstrating the high reliability of the survey items in this study.

4.4. Discussion

In this study, the understanding of the COLREGs sailing rule applied to MASSs was investigated and analyzed, thereby verifying the following study achievements, from which the following discussion issues were derived.

4.4.1. Problem to apply the current COLREGs for collision avoidance of MASSs

As discussed earlier, the most recently reviewed studies developed and applied collision avoidance algorithm of MASSs based on COLREGs. The COLREGs applied in recently reviewed studies is crucial as it determines the standards that set the relationship of rights and duties in the navigation of ships. The relationship between the give-way and stand-on vessels is determined using COLREGs and the proportion of mistakes is calculated based on whether the sailing rules are complied with or not when a collision accident occurs.

However, COLREGs exhibits several theoretical limitations when applying navigations as follows:

First, COLREGs has a problem with respect to the timing to apply the sailing rules. A sailing rule to be applied between two ships may be different according to the timing of the sailing rule application when a collision accident of ships occurs. COLREGs does not thoroughly specify the definitions or requirements of the timing to be applied the sailing rules. In fact, when a ship collision accident occurs, maritime law courts around the world determine the timing of applying the sailing rule case by case considering the circumstance at the time of the collision, navigator's knowledge, customs, precedents, theories, etc. Thus, the timing of applying the sailing rules for both ships should be specified or interpreted more clearly when determining the collision avoidance model for MASSs. However, because detailed and sufficient data regarding customs, precedents, etc. of such cases are not collected, the discussion of the application timing in this study will be conducted through additional research on the timing to apply the sailing rules for MASSs in the future.

Second, COLREGs has a problem with the concept of navigation and practical application of navigation. For example, the physical concept of overtaking ship means a ship overtaking when coming up with another vessel from a direction of $> 22.5^\circ$ abaft its beam, which refers to the following ship. Based on this concept, numerous studies have been conducted. However, understanding compass bearing is a physical method, which is difficult for navigators to apply in real situations. Thus, the use of the visible side light method is more practical than using the compass bearing method from the navigator's viewpoint. This indicates

that the overtaking ship is considered as the following ship that attempts to overtake from the location where the side light of the leading ship is not seen but only the stern light is seen. However, care should be taken because the visible distance of the stern light may vary depending on the size and visibility of the ship. In addition, when the position of the following ship is not certain with respect to overtaking or crossing relations, COLREGs stipulates that this situation shall be recognized as the overtaking ship, which is vague to navigators. Thus, we aimed to configure a collision avoidance model based on the radar interpretation, which was widely used to determine the collision risk by navigators, in contrast with recently reviewed studies, and verify the understanding of applying collision prevention by navigators.

Third, COLREGs has a problem of wide range to apply crossing sailing rules. Crossing states are the most frequently occurring form in navigation situations of marine traffic. In the mutual relationship between course and speed of two ships, if the bearing of the ship (stand-on ship), which has the other ship on the starboard side, does not change, the two ships will meet at the crossing point at the same time, resulting in a collision. This is the most frequent cause of the collision. The range of crossing relations is considerably wide as it includes all situations excluding head-on or overtaking cases of navigating power-driven vessels. If we understand this situation in terms of physical concept, it refers to all crossing navigation relations except for cases when facing the head each other in the visible range of the masthead light or side light. Therefore, crossing situations are different depending on the relationship between the ships. Hence, it is difficult to determine the risk of collision and how to avoid it accordingly. Thus, this study aimed to verify the understanding of the sailing rule interpretation of COLREGs based on the wide range through the survey.

4.4.2. Implications of the survey results

In this study, a collision avoidance situations was designed based on radars to verify the understanding of sailing rule interpretation of COLREGs as discussed earlier and a survey was conducted based on the situation to derive the following implications. Basically, we used a standard deviation to verify the misunderstanding.

- Implications of the survey results concerning the misunderstanding in head-on and crossing situations (HC_Q1 (000°)–HC_Q11 (020°)).

As presented in Appendix 4, as the relative bearing increased, the proportion of recognizing it as a crossing state by navigators tended to increase, which indicated that navigators were well aware of the situations in the survey. The misunderstanding was not considerably large up to HC_Q3 (004°) but started to increase from HC_Q4 (006°) and peaked at HC_Q6 (010°); subsequently, the misunderstanding exhibited a decreasing trend. Thus, because the navigators did not see misunderstanding up to a relative bearing of 004° but vagueness peaked at 010° , they had the least confidence whether a head-on or crossing situation should be applied to the navigation with the ship with a relative bearing of 010° .

- Implications of the survey results concerning the misunderstanding in OC situations (OC_Q1 (180°)–OC_Q10 (090°)).

As presented in Appendix 4, as the relative bearing increased, the proportion of recognizing it as a crossing state by navigators tended to increase, which indicated that navigators were well aware of the situations in the survey. The misunderstanding was not considerably large up to OC_Q2 (170°) but started to increase from OC_Q3 (160°) until OC_Q10 (090°), showing an increasing trend. Thus, navigators did not show misunderstanding up to a relative bearing of 170° and then started to be less confident whether an overtaking or crossing situation should be applied to the navigation until 90° .

- Implications of the survey results.

As verified in Appendix 4, standard deviations in the range of HC_Q3 (004°) to HC_Q4 (006°) for the misunderstanding of head-on and crossing situations and the range of OC_Q2 (170°) to OC_Q3 (160°) for the misunderstanding of OC increased rapidly. These results indicated that the range of confidence in navigation application by navigators was

narrow and with the increase in misunderstanding, the tendency of avoiding the risk of collision by navigators became strong by taking an active rather than passive collision avoidance action. In addition, the standard deviation of junior officers was relatively larger than that of senior officers, which indicated that the tendency of avoiding collision is greater in junior officers than in senior officers.

4.4.3. Navigator survey and collision avoidance algorithm for MASS

The results of the navigators' survey were compared with the results used in the algorithm for MASS. First, the HC situation, as the angle increased at 008°, more cases were interpreted as a crossing situation. However, algorithm designers understand that HC situations encompass a range of angles, approximately from 0° to 45°. The application of an "average" to the research was challenging, and the majority of researchers and mariners considered angles within approximately 10° as HC situations.

In addition, in case of the OC situation, although COLREG clearly defines the overtaking situation, the rate of non-response to crossing or overtaking increased as it moved from 130° to a small angle as a result of the navigators' survey. This means that there is a problem in the understanding of the navigator and the algorithm designer, and a way to solve it will be needed.

4.4.4. Contributions of the study

Based on the aforementioned analysis, the contributions of the study are as follows:

First, we verified that navigators tended to interpret Sectors A and B more narrowly those described by the rules. This tendency will provide a basis to navigators for performing an active avoidance action rather than a passive avoidance action in a vessel collision situation. Moreover, this tendency will continue in the collision avoidance action between manned ships and MASSs. Thus, this tendency is necessary for the development of a collision avoidance algorithm with MASSs for further extending a sector of active collision avoidance actions performed by manned ships.

Second, navigators with fewer years of experience tended to take more active collision avoidance actions. Thus, the development of the collision-avoidance model for MASSs must extend and apply a sector of collision avoidance actions in the navigational watch time.

Third, this study suggests the need for a discussion on several issues described in the following sections.

4.4.5. Sailing rules

This study confirms the need to discuss the theoretical limitations of navigation analysis. In particular, it highlights the necessity of discussing when to apply sailing rules. IMO recommended temporary guidelines for MASS trial operations during its 101st meeting in 2021, and at its 106th meeting, a draft was prepared for the development of the MASS code starting in 2022. The current timeline for MASS code development plans to approve and implement the nonmandatory MASS code in 2024, adopt the mandatory MASS code in 2025, and enforce the mandatory MASS code in 2028[31]. During the 107th MSC (Maritime Safety Committee) meeting in 2023, discussions were held regarding the definition of MASS, the main principles governing MASS, the terminology for remote control centers, and the qualification requirements for captains and crew [44]. Based on the outcomes of the MSC meeting, it was collectively agreed that there is no immediate need to revise the COLREGs agreement for the operation of MASS. Furthermore, discussions regarding the revision of Standards of Training, Certification and Watchkeeping for Seafarers were deemed premature at this stage [44].

Notably, several studies have been initiated to explore the necessity of revising COLREGs in response to the challenges related to developing measures for avoiding ship collisions [35,37,47,52,54,64] although no revisions are currently included in the MASS code. In the long term, any revision of the COLREGs agreement should focus on discussing the relationships between the existing sailing and autonomous ships,

power-driven vessels and autonomous ships, and other relevant aspects rather than undertaking extensive and immediate revisions.

The content related to the MASS principle was not included in discussions about how to navigate in the event of encounters between ships [32]. This omission is likely a contributing factor to the difficulty in finding a simple solution to such problems. To address the challenges of managing ship-to-ship encounters and ship collision scenarios, it is essential to delve into the interpretation of the qualitative rules of COLREGs and the operational principles of autonomous ships. For instance, we can draw parallels between the emergence of autonomous ships during the current Fourth Industrial Revolution and the development of power-driven vessels during past industrial revolutions. This comparison can provide insights into the evolving principles of ship operation throughout maritime history. It is important to examine how the operational principles of sailing ships from the past compare with those of power-driven vessels and how the operating principles of contemporary autonomous ships align with those of power-driven vessels.

Furthermore, for MASSs that do not follow the high-speed ships and lights visually, a new concept of the timing to apply sailing rules is needed to be discussed instead of applying the timing based on the visible-light range in the COLREGs. However, currently, references such as precedents and research cases are not sufficient compared with those in case of manned ships concerning the timing of applying sailing rules. Thus, it is essential to thoroughly analyze the aforementioned issue in future studies.

5. Conclusions

This study aims to clarify the collision avoidance situations by identifying the misunderstanding in the sailing rule interpretation of COLREGs. According to COLREGs, sailing, overtaking, head-on, and crossing situations are applied to collision avoidance situations and the ship's action of collision avoidance varies according to the sailing rule interpretation. Thus, we conducted a survey to identify the understanding of navigators, compared the result of navigator's survey, and reviewed the collision-avoidance algorithm used in recent studies. In many recent studies, the range from 0° (front) to 20° has been applied in HC situations and the OC situations ranges were set at 90–180° as per Appendix 1.

- Navigators were not sure about which sailing rule should be applied in case of HC situations for ships coming from 008° from the head.
- Navigators tended to apply a crossing situation rule for ships coming from 014° to 090° from the head but the standard deviation was large, revealing the misunderstanding of the sailing rule interpretation.
- In OC situations, as the angle increases from 130°, analysis has confirmed that it is difficult to understand the situation and that navigators have difficulty interpreting sailing rules.

In conclusion, a notable disparity emerged between the sailing rule based on COLREGs and the automated collision avoidance algorithms. Moreover, it was substantiated that practical navigators exhibited divergent interpretations of these regulations.

Based on the results of this study, the timing of applying the sailing rules of COLREGs should be more clearly stipulated or interpreted. In accordance with the forthcoming MASS code, it is imperative to engage in a thorough discussion regarding safety issues that could potentially manifest across various sailing situations concurrently. Furthermore, because the designations of stand-on and give-way ships may be different depending on the relationship of crossing ships in the crossing navigation situation, more important issues may arise. Thus, additional studies are needed to solve this problem.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that has been used is confidential.

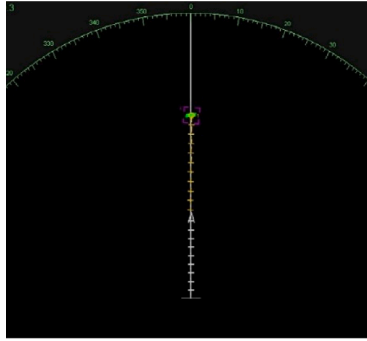
Acknowledgments

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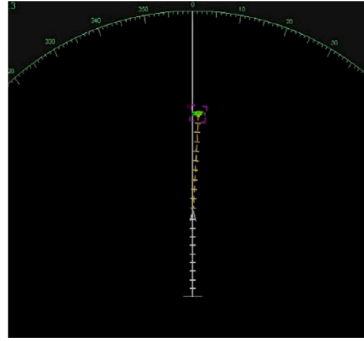
Appendix 1. Related works regarding collision avoidance of a ship applying with COLREGs

Related Works	Angle of Head-on Situation (°) (Sector A)		Angle of Crossing Situation (°) (Sector B)	Angle of Overtaking Situation (°) (Sector C)
[39]	-		107.5	135
[51]	-		107.5	135
[63]	-		107.5	135
[18]	-		112.5	135
[62]	5		132.5	90
[10]	10		85	180
[25]	10		107.5	135
[38]	10		107.5	135
[89]	10		107.5	135
[41]	10		105.5	135
[57]	10		107.5	135
[65]	10		130	90
[82]	10		130	90
[86]	10		107.5	135
[87]	10		107.5	135
[91]	10		107.5	135
[92]	10		130	90
[22]	11.4		106.8	135
[7]	12		119	110
[46]	12		119	135
[17]	12		106.5	135
[61,93]	12		106.5	135
[81]	12		106.5	135
[23]	12		168	12
[85]	15		105	135
[5]	20		160	20
	20			
[9]	20		160	20
[90]	10		107.5	135
[42]	20		160	20
[76]	20		160	20
[92]	20		102.5	135
[34]	22.5		101.25	135
[3]	30		97.5	135
-	30		97.5	135
[49]	30		97.5	135
[72]	30		97.5	135
[14]	45		132.5	50
[20]	45		132.5	50
[59]	45	90		135
[66]	50	87.5		135
[95]	50	87.5		135
[21]	120	90		60
[43]	120	90		60
[40]	-	-		-
[15]	-	-		-
[88]	-	-		-
[27]	-	-		-
[78]	-	-		-
[45,50]	-	-		-

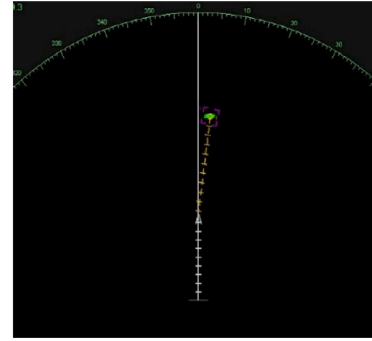
Appendix 2. Survey of collision avoidance situation with navigators between head-on and crossing (HC) situation



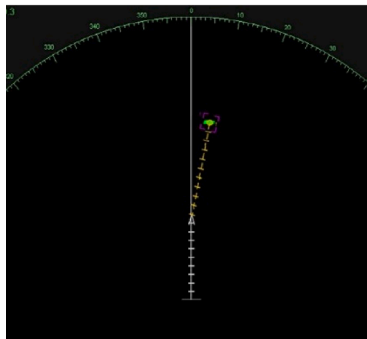
HC_Q1) Aspect-000°



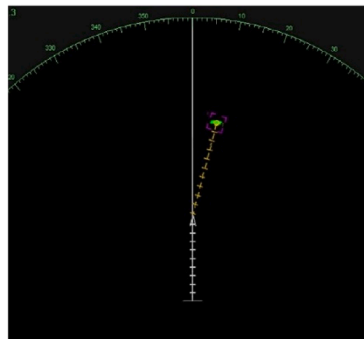
HC_Q2) Aspect-002°



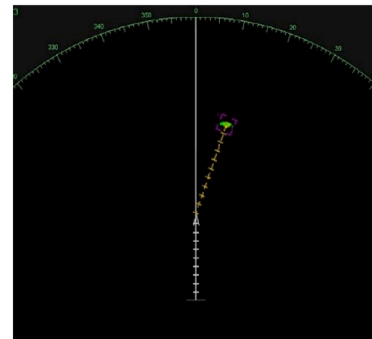
HC_Q3) Aspect-004°



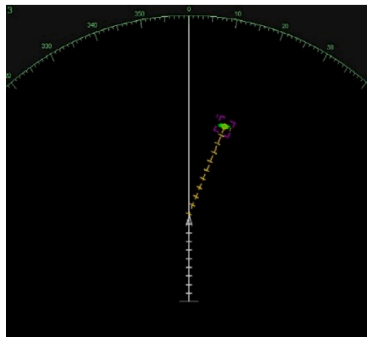
HC_Q4) Aspect-006°



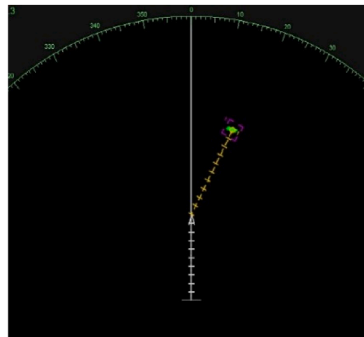
HC_Q5) Aspect-008°



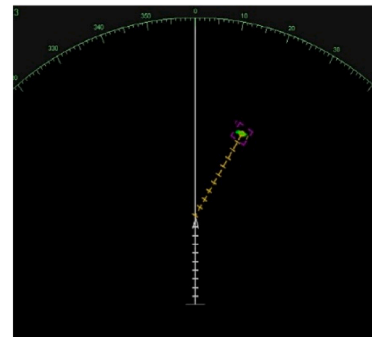
HC_Q6) Aspect-010°



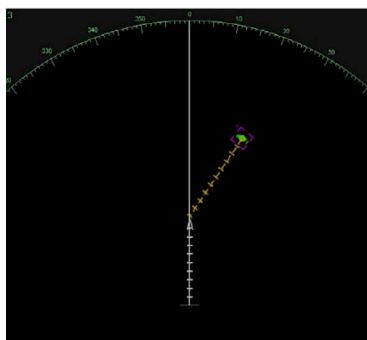
HC_Q7) Aspect-012°



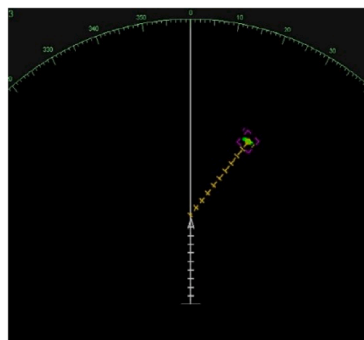
HC_Q8) Aspect-014°



HC_Q9) Aspect-016°

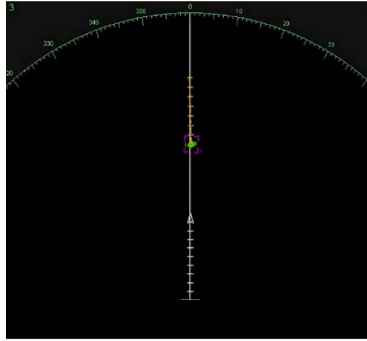


HC_Q10) Aspect-018°

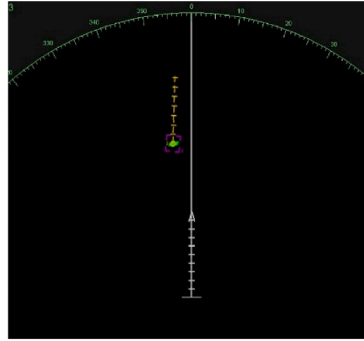


HC_Q11) Aspect-020°

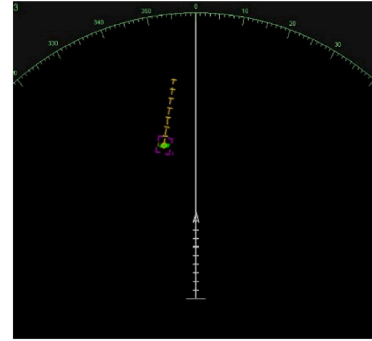
Appendix 3. Survey of collision avoidance situations with navigators between overtaking and crossing (OC) situation



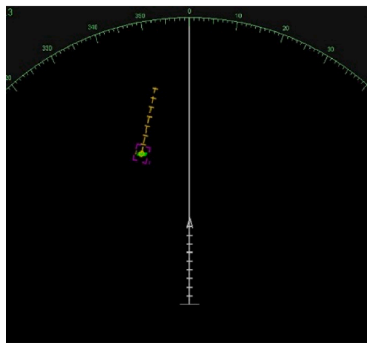
OC_Q1) Aspect-180°



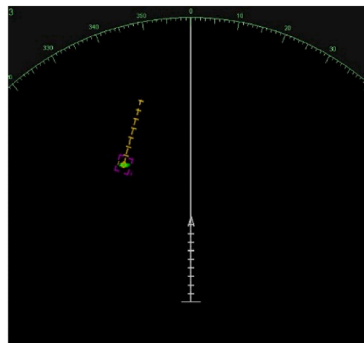
OC_Q2) Aspect-170°



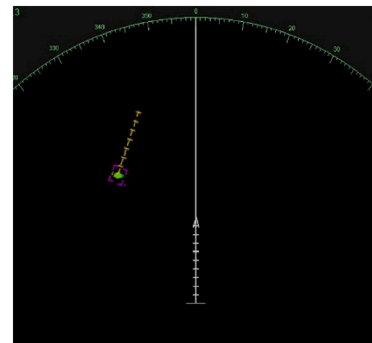
OC_Q3) Aspect-160°



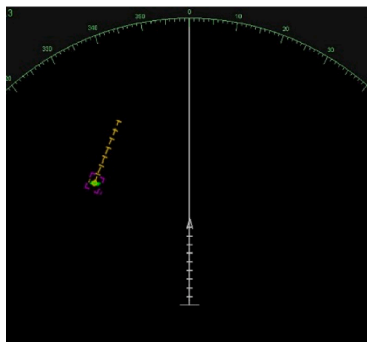
OC_Q4) Aspect-150°



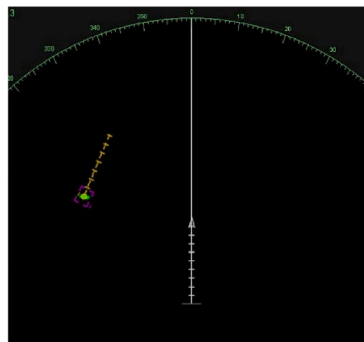
OC_Q5) Aspect-140°



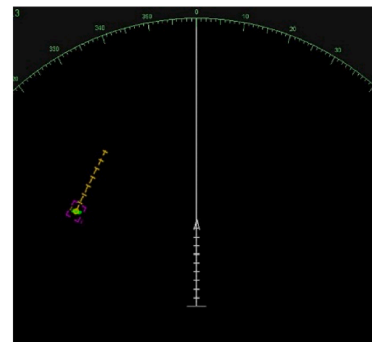
OC_Q6) Aspect-130°



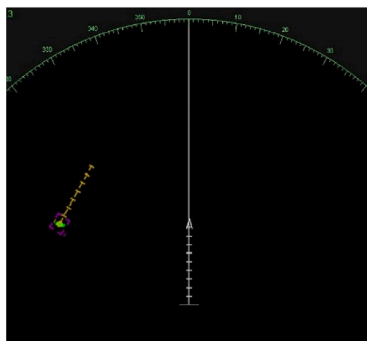
OC_Q7) Aspect-120°



OC_Q8) Aspect-110°



OC_Q9) Aspect-100°



OC_Q10) Aspect-090°

Appendix 4. Results of responses in the survey questionnaire to identify navigator's understanding of collision avoidance situations

Number of Question (°)	Number of Response 1	Number of Response 2	Number of Response 3	Number of Response 4	Number of Response 5	Average (Standard Deviation)
HC_Q1 (000)	92	9	0	0	0	1.09(0.29)
HC_Q2 (002)	79	22	0	0	0	1.22(0.41)
HC_Q3 (004)	58	42	1	0	0	1.44(0.52)
HC_Q4 (006)	40	44	1	13	3	1.96(1.09)
HC_Q5 (008)	30	46	1	17	7	2.26(1.25)
HC_Q6 (010)	23	32	2	30	14	2.80(1.44)
HC_Q7 (012)	17	29	1	40	14	3.05(1.39)
HC_Q8 (014)	14	17	4	41	25	3.46(1.39)
HC_Q9 (016)	12	15	4	35	35	3.65(1.40)
HC_Q10 (018)	8	9	4	41	39	3.93(1.23)
HC_Q11 (020)	9	9	2	31	50	4.03(1.30)
OC_Q1 (180)	73	20	8	0	0	1.36(0.63)
OC_Q2 (170)	69	27	5	0	0	1.37(0.58)
OC_Q3 (160)	32	47	6	11	5	2.11(1.12)
OC_Q4 (150)	27	47	2	18	7	2.32(1.24)
OC_Q5 (140)	19	50	4	21	7	2.48(1.21)
OC_Q6 (130)	13	42	6	30	10	2.82(1.27)
OC_Q7 (120)	14	42	8	24	13	2.80(1.30)
OC_Q8 (110)	10	44	7	22	18	2.94(1.33)
OC_Q9 (100)	11	33	14	23	20	3.08(1.34)
OC_Q10 (90)	8	31	8	29	25	3.32(1.35)

Appendix 5. General statistics of responses in the survey questionnaire to identify navigator's understanding of collision avoidance situations

Age		Vessel tonnage		Position		Experience	
Twenties	25	less than 5000 tons	10	Captain	7	> 1 year	4
Thirties	54	5000–10,000tons	13	Chief mate	35	1–3 years	29
Forties	17	10,000–50,000tons	29	Second mate	44	3–5 years	27
Fifties	4	50,000–100,000	33	Third mate	14	5–10 years	29
Sixties	1	More than 100,000 tons	16	Pilot	1	10 years<	12
Total	101		101		101		101

Appendix 6. Factor Analysis result of navigators' understanding of collision situations survey for HC_Q1 to HC_Q11

Item	1	2	3	4
HC_Q5 (008)	.863	.196	.248	.167
HC_Q4 (006)	.769	.168	.111	.380
HC_Q6 (010)	.722	.264	.403	.195
HC_Q7 (012)	.585	.335	.528	.106
HC_Q11 (020)	.195	.909	.295	-.078
HC_Q10 (018)	.226	.892	.303	-.049
HC_Q8 (014)	.354	.473	.771	-.009
HC_Q9 (016)	.294	.524	.758	-.018
HC_Q2 (002)	.086	-.064	.006	.761
HC_Q3 (004)	.337	.029	.109	.687
HC_Q1 (000)	.115	-.058	-.078	.539
Eigen value	2.634	2.376	1.880	1.571
Common Variance(%)	23.950	21.604	17.094	14.280
Cumulative Variance(%)	23.950	45.553	62.647	76.927

KMO = .843, Bartlett's X² = 962.212(p < .001)**Appendix 7. Factor Analysis result of navigators' understanding of collision situations survey for OC_Q1 to OC_Q10**

Item	1	2	3
OC_Q9 (100)	.907	.159	-.130
OC_Q8 (110)	.850	.285	.009
OC_Q7 (120)	.834	.257	-.046
OC_Q10 (90)	.752	.182	-.194
OC_Q4 (150)	.247	.867	.098
OC_Q3 (160)	.061	.773	.297
OC_Q5 (140)	.356	.757	.196
OC_Q6 (130)	.492	.714	.002

(continued on next page)

(continued)

Item	1	2	3
OC_Q2 (170)	-.091	.238	.867
OC_Q1 (180)	-.117	.112	.829
Eigen value	2.634	2.376	1.880
Common Variance(%)	32.631	27.080	16.324
Cumulative Variance(%)	32.631	59.711	76.035

KMO = .859, Bartlett's X² = 716.169(p < .001)

Appendix 8. Reliability Analysis result of navigators' understanding of collision situations survey

Item	Cronbach's alpha	N
HC_Q1~11	.903	11
OC_Q1~10	.873	10

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